

Privacy-Evidence-Oriented Monitor/Obvserver Synthesis

Privacy/Evidence Proofs
Towards formally independent & evident logs

Rehab Massoud, 16.09.2026

Motivation

- During Postmortem?

Why Investigators Fear the Two Boeing 737s Crashed for Similar Reasons

By JAMES GLANZ, K.K. REBECCA LAI and JIN WU | MARCH 13, 2019

American and Canadian regulators cited newly available satellite-tracking data in [grounding all Boeing 737 Max jets](#) on Wednesday, saying there were similarities between the deadly crash involving the plane in Ethiopia on Sunday and a Indonesian flight in October.

The causes of both crashes are still under investigation. But the data cited by regulators points to preliminary indications that the two planes could have been brought down by the same cause, a malfunctioning automated system intended to keep the jet from stalling.



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Logs from System vs Driver/Witness?

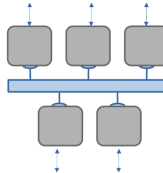
System logs

```
2.257008s GearBoxInfo(1020)d 1 01          size -> 58
2.253552s EngineData(100)d 8 00 00 19 00 00 00 00 00-> 125
2.256312s ABSdata(201)  d 6 00 00 00 00 00 00 00    -> 105
2.260804s Ignition_Info(103) d 2 01 00          -> 67
...
```

Driver Claim

It was unintended
acceleration

Catch me
if you can



Catch me
if you can

- How?
- Time as the main traced artifact ?
- ***“Change over Time”*** as the main traced artifact
- Periodic logging, trace-cycle based *“lossy”* compression
- We’re not spying nor gathering info, i.e. system info is available, use it!
- Focus on logging evidence for what needs to be proven

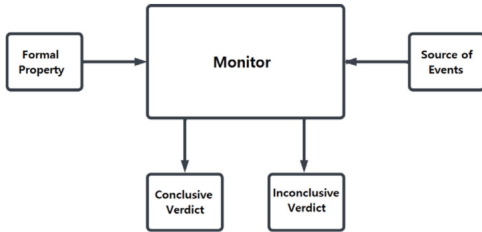
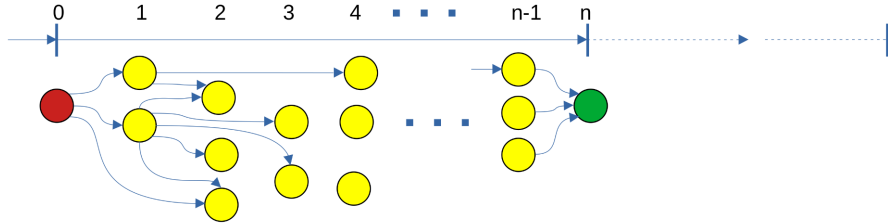


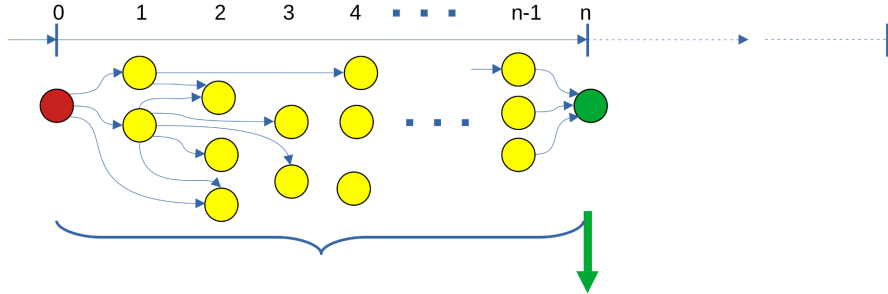
Fig. 1. RV Setup.

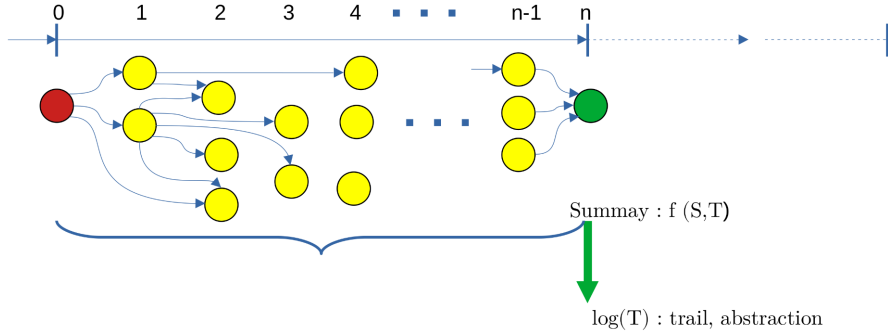


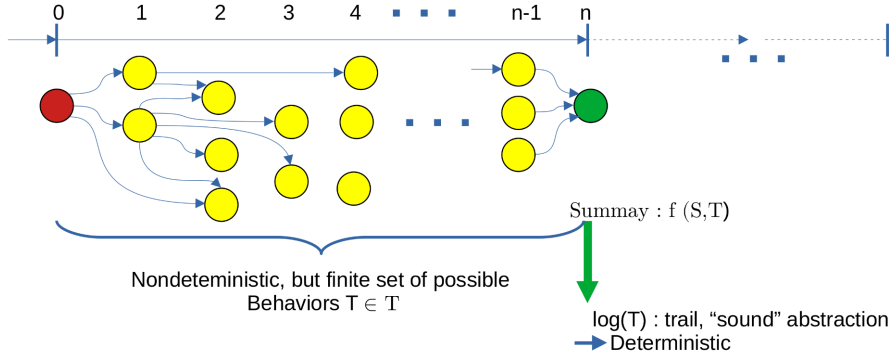
Fig. 2. Steps required to generate an FSM from an LTL formula φ .

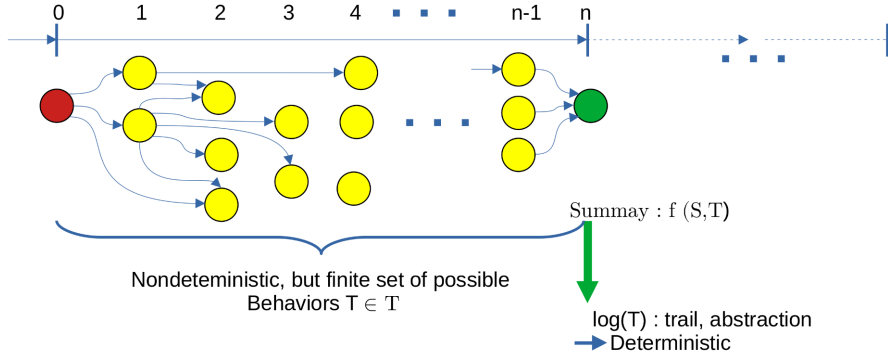
Privacy/Evidence-Oriented Monitor/Obvserver Synthesis and Privacy/Evidence Proofs



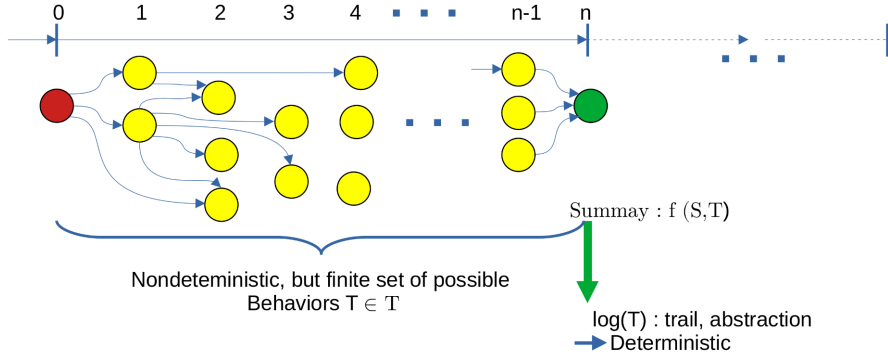






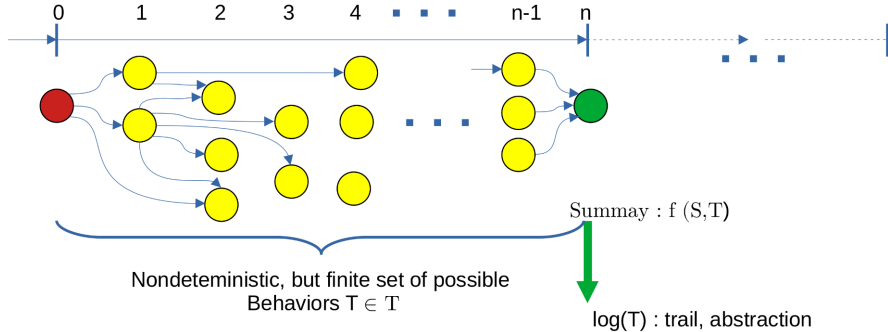


Question 1: Given $\log(T)$, a set T and a Property $P \subset T$
can $\log(T) \models (T \in P)$? or can $\log(T) \models (T \in T \setminus P)$?



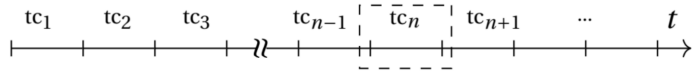
Question 2: Given $\log(T)$, a set T and a Property $P \subset T$

is it the case that $\forall T \in \mathcal{T}$, either $\log(T) \models (T \in P)$ or $\log(T) \models (T \in \mathcal{T} \setminus P)$? i.e. never both?



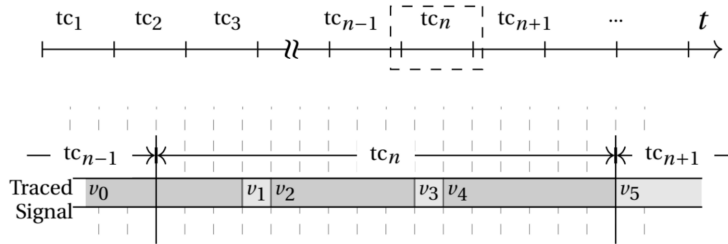
Question 3: Can we find $f(S, T)$, such that the resulting $\log(T)$, a set $\mathcal{T}$ and a Property $P \subset \mathcal{T}$ is it the case that $\forall T \in \mathcal{T}$, either $\log(T) \models (T \in P)$ or $\log(T) \models (T \in \mathcal{T} \setminus P)$? i.e. never both?

Case study: Application

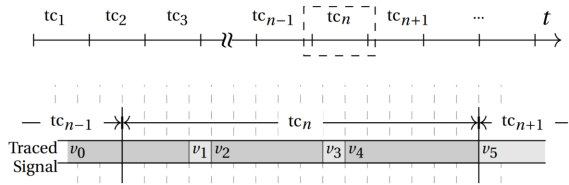


- [1] R. Massoud et al, "Temporal Tracing of On-Chip Signals using Timeprints", Design Automation Conference, DAC 2019.
- [2] R. Massoud, "Evidence-Oriented Tracing and Verification, The declaration of Timeprints", PhD Thesis, University of Bremen, 2021.
- [3] R. Massoud, H. M. Le, R. Drechsler, "Property-driven Timestamps Encoding for Timeprints-based Tracing and Monitoring", Proceedings of the International Conference on Formal Modeling and Analysis of Timed Systems, Pages 41-58, (FORMATS) 2019.

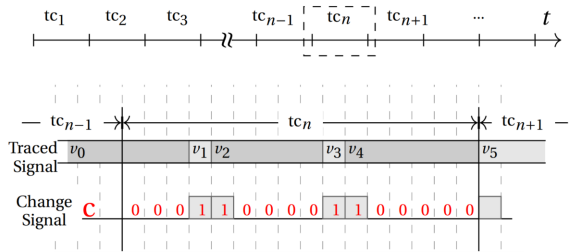
Case study: Application



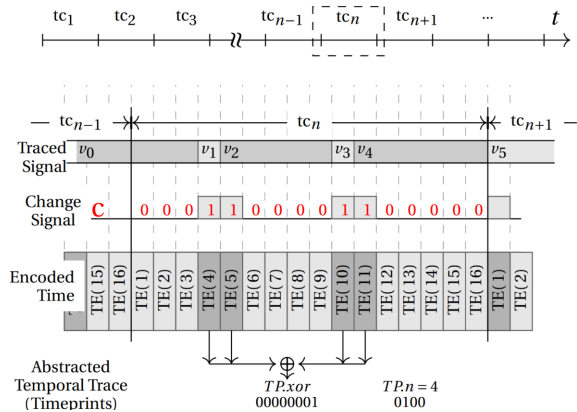
Timeprints and Trace- Cycles (Example)

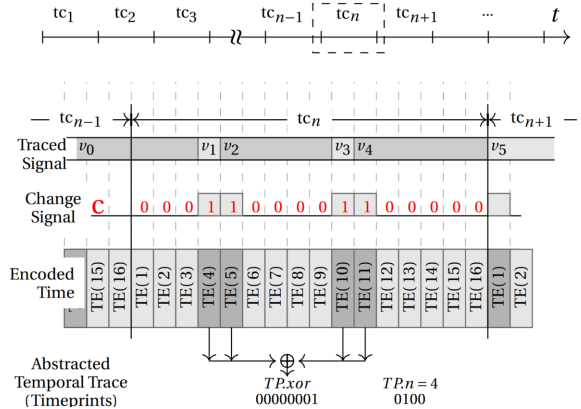
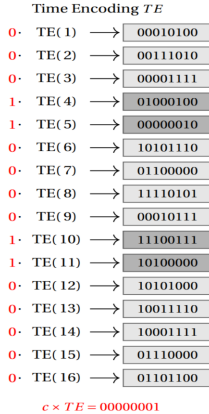


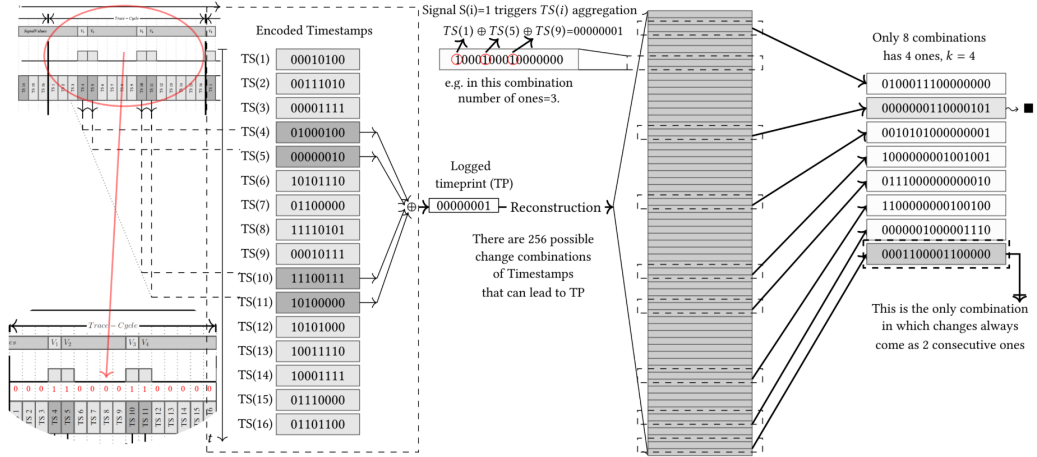
Timeprints and Trace- Cycles (Example)



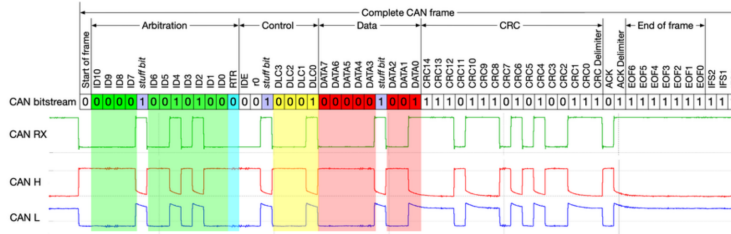
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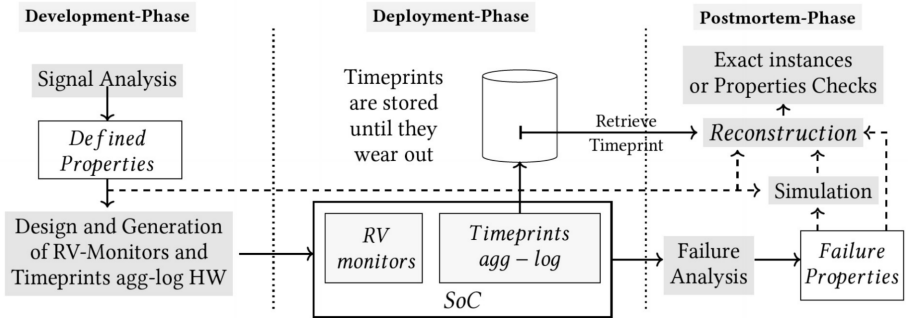
CAN-Bus messages delay (Evidence)



For a 1 Mbps CAN Bus, using 1000 clock-cycles long trace-cycle of length one ms, the Timeprints log is 34(24+10) Kbits. Longer trace-cycles are possible.

Ideas

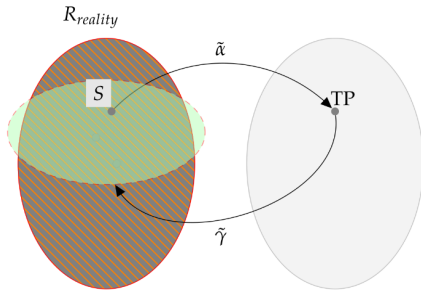
- Proof structures that facilitate human-checking and oversight of evidence
- Theoretical proofs of the limits of multi periodic-lossy-abstraction/logs for providing evidence
- Proving limits of observation/monitoring of temporal properties/events with regard to their (expressiveness, cardinalities, intervals, ...)



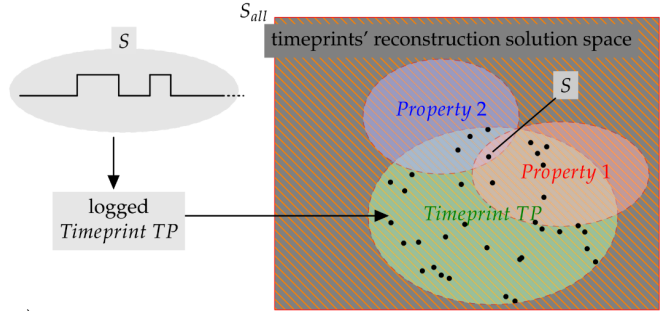
RV: Run-Time Verification

Timeprints agg-log: Timeprints aggregation and logging module

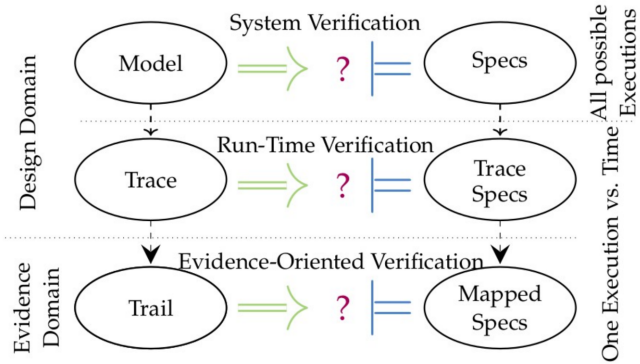
R. Massoud et al, "Temporal Tracing of On-Chip Signals using Timeprints", Design Automation Conference, DAC 20129.



$R_{reality}$ is the set of all signals S (all possible changes) that can happen in reality, one point S actually happened



S_{all} is the set of all possible reconstructions from all timeprints
 $R_{reality} \subset S_{all}$



R. Massoud, "Evidence-Oriented Tracing and Verification, The declaration of Timeprints", PhD Thesis, University of Bremen, Dec. 2021.

Tesla told France there was no sign of technical fault in Paris crash

By **Mathieu Rosemain** and **Elizabeth Pineau**

December 16, 2021 12:41 AM GMT · Updated December 16, 2021



[1/2] A Tesla logo is seen on a Parisian taxi car in Paris, France, December 14, 2021. REUTERS/Sarah Meysonnier/File Photo [Purchase Licensing Rights](#) 



Summary Companies

- One killed, 20 injured, three seriously, source says

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World ▾ Business ▾ Markets ▾ Sustainability ▾ Legal ▾ Commentary ▾ Technology ▾ Investigations ▾

Ricordel cited Tesla as saying on Monday that an initial inquiry had ruled out a technical dysfunction of the vehicle. The G7 executive said the driver had tried to brake but the car instead accelerated. It was unclear if the car was operating in Autopilot mode.

Jerome Coumet, mayor of Paris' 13th arrondissement, said on Twitter, "The first elements of the investigation indicated that the accelerator would have gotten stuck."

Ricordel said: "Today, we have two divergent views on the subject. We will maintain the suspension of Tesla Model 3 while the investigation is ongoing, as a safety measure for our drivers, customers and other road users."

Tesla collects detailed data from the sensors and cameras on its vehicles and has used such data in the past to challenge claims that accidents were caused by malfunctioning technology.

